

Problem-Solving Exercises

Coding Theory

Question 1

We propose, throughout this question, to determine the expression of the capacity of the communication channel defined in Figure 1.

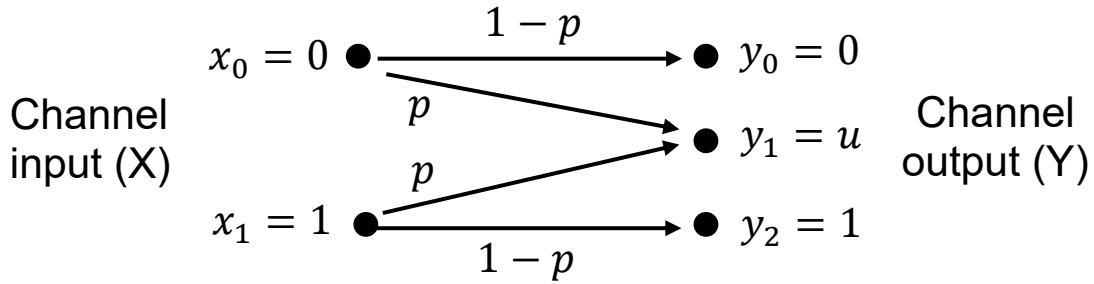


Figure 1

The channel input is a binary source X that can generate a symbol $x_0 = 0$ with probability $\Pr\{x_0\} = 0.5$ or a symbol $x_1 = 1$ with probability $\Pr\{x_1\} = 0.5$.

The channel output is a source Y that can generate three symbols as follows: $y_0 = 0$ with probability $\Pr\{y_0\}$, $y_1 = u$ (as in 'unknown') with probability $\Pr\{y_1\}$, or a symbol $y_2 = 1$ with probability $\Pr\{y_2\}$.

The parameter p denotes the probability that a transmitted symbol will be erased during transmission over the channel.

The capacity of the channel can be determined using the generic expression

$$C = E \left\{ \log_2 \left(\frac{\Pr\{y_j|x_i\}}{\Pr\{y_j\}} \right) \right\} = \sum_{i=0}^1 \sum_{j=0}^2 \Pr\{x_i \cap y_j\} \cdot \log_2 \left(\frac{\Pr\{y_j|x_i\}}{\Pr\{y_j\}} \right).$$

1.a Determine the expressions of the three probabilities $\Pr\{y_j\}$, $j \in \{0, 1, 2\}$.

[5 marks]

1.b Use the results obtained in Question 1.a to determine the expression of the capacity C of the channel described in Figure 1.

[10 marks]

Question 2

All linear block codes can be defined using a parity-check matrix. In other words, the operation of any linear block code can be described using a Tanner graph that connects a set of bit nodes to another set of check nodes.

In this question, assume transmission over a binary symmetric channel (BSC) with an error probability p .

Also assume that the bit-flipping algorithm is used to decode a particular linear block code over this BSC.

We have seen in class that the bit-flipping algorithm is going to work well provided that each check node in the Tanner graph of this linear block code is fed with, at most, one wrong bit.

2.a Consider a particular check node connected to l bit nodes $c_0, c_1, c_2, \dots$, and c_{l-1} .

Find the expression of the probability that this check node receives two or more wrong bits.

Simplify the equation thus obtained by assuming that $p \ll 1$.

[10 marks]

2.b By using the result obtained in Question 2.a, determine a condition that must be satisfied for successful decoding of a linear block code using the bit-flipping algorithm.

[10 marks]

Question 3

Consider the rate-1/3 turbo encoder depicted in Figure 2. It is designed by parallel concatenation of two recursive and systematic convolutional (RSC) codes separated by a random interleaving function.

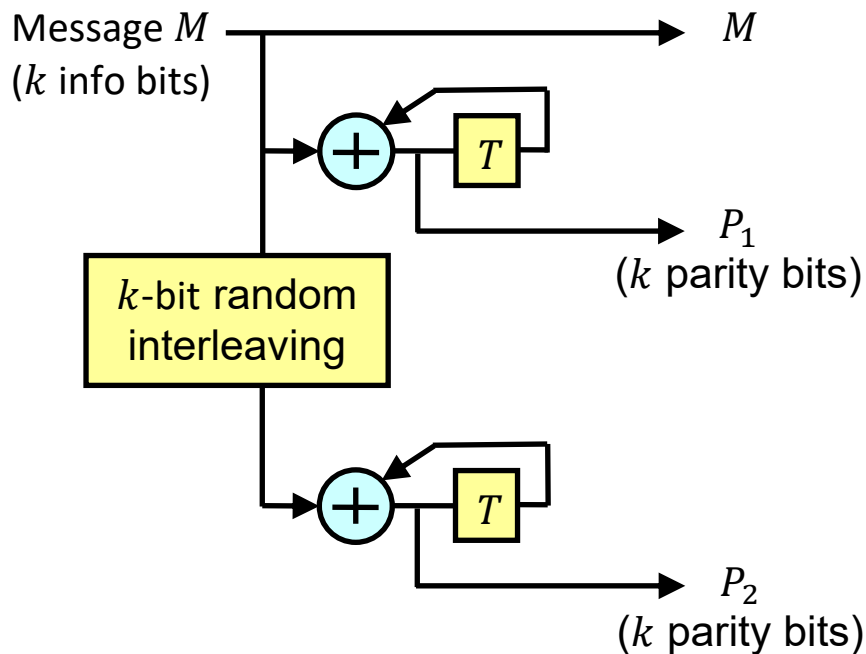


Figure 2

3.a Draw the trellis diagram for the first RSC encoder which is the encoder generating the first part, M and P_1 , of a codeword.

Also, draw the trellis diagram for the second RSC encoder which is the encoder generating the second part, P_2 , of a codeword.

[5 marks]

3.b Determine the first three terms in the union bound of the turbo encoder shown in Figure 2.

Is it a good turbo code?

[10 marks]

3.c We have seen in class that turbo codes are decoded using an iterative decoding algorithm.

This algorithm can perform very close to the maximum-likelihood decoding. How do we know that? In other words, what is the evidence behind that statement?

[5 marks]

Question 4

Figure 3 shows the capacity limit of a BPSK, AWGN channel as a function of the SNR per symbol, $\frac{E_s}{N_0}$.

Consider a communication system using BPSK modulation and a channel code with a coding rate R_c to transmit information over an AWGN channel.

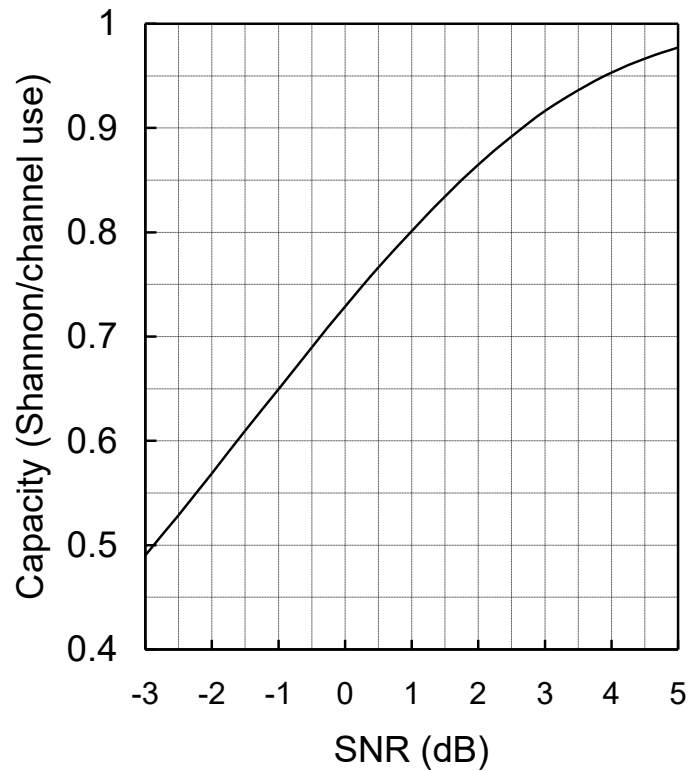


Figure 3

Fill in the table below by using the plot shown in Figure 3.

Coding rate R_c	Minimum SNR per Shannon, $\frac{E_b}{N_0}$, in dB, required to achieve error-free communications
0.5	
0.6	
0.7	
0.8	
0.9	

[10 marks]

Question 5

We propose to design a channel code, called $C_{1,2}$, by combining two linear block codes C_1 and C_2 .

The parameters of C_1 are k_1 , n_1 , and $d_{min,1}$, whereas the parameters of C_2 are k_2 , n_2 , and $d_{min,2}$, where k_1 and k_2 denote the message lengths, n_1 and n_2 designate the codeword lengths, and $d_{min,1}$ and $d_{min,2}$ are the minimum Hamming distances between codewords.

The encoding algorithm is explained below.

Each message composed of $k_1 \times k_2$ information bits is stored in a two-dimensional (2-D) array of size $k_1 \times k_2$ bits. This array thus has k_1 columns and k_2 rows.

At the first step of encoding, each row of this array is encoded into a codeword of length n_1 bits using C_1 . The row encoding results in a 2-D array composed of n_1 columns and k_2 rows.

At the second and final step of encoding, each of the n_1 columns is encoded into a codeword of length n_2 using C_2 . The column encoding results in a 2-D array composed of n_1 columns and n_2 rows. This array is the codeword associated with the $k_1 \times k_2$ -bit array present at the encoder output.

5.a The minimum Hamming distance, d_{min} , between codewords for $C_{1,2}$ is the product of $d_{min,1}$ and $d_{min,2}$. In other words, we can write $d_{min} = d_{min,1} \cdot d_{min,2}$.

How would you demonstrate this result?

[10 marks]

5.b How would you decode the code $C_{1,2}$ in practice?

To illustrate your explanations, feel free to draw the decoder structure that you propose.

[10 marks]

Question 6

Propose a method for determining the minimal distance, d_{min} , of a linear block code by only using its Tanner graph representation.

Illustrate your method by determining the minimal distances for the codes defined by the Tanner graphs shown in Figures 4 to 7.

Figure 4

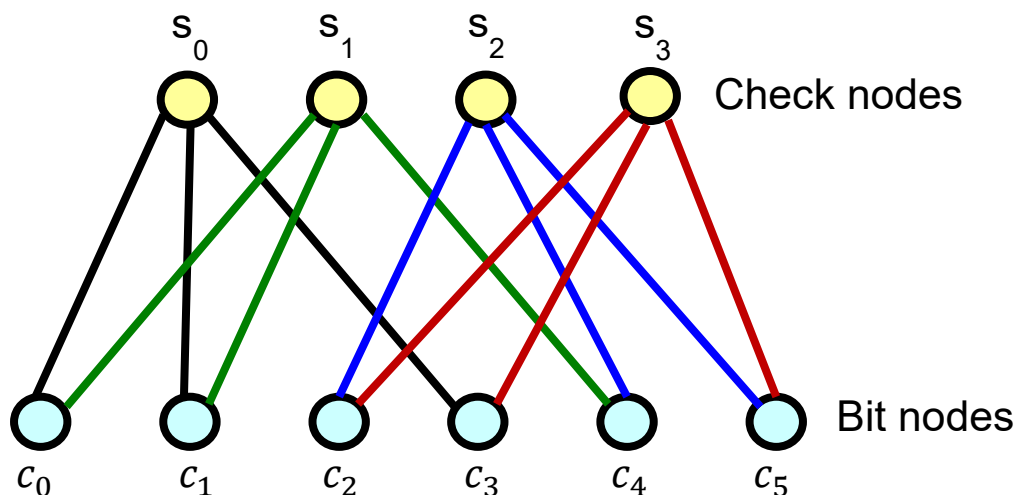


Figure 5

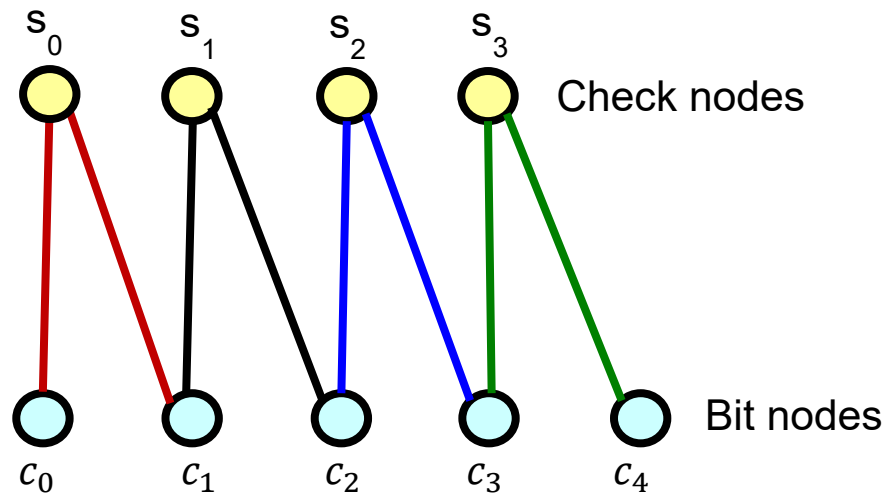


Figure 6

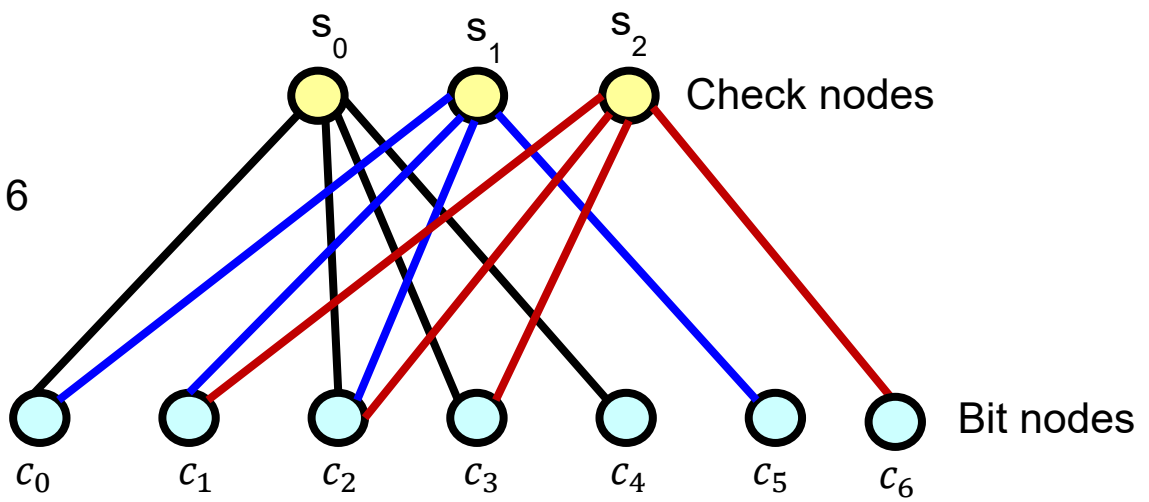
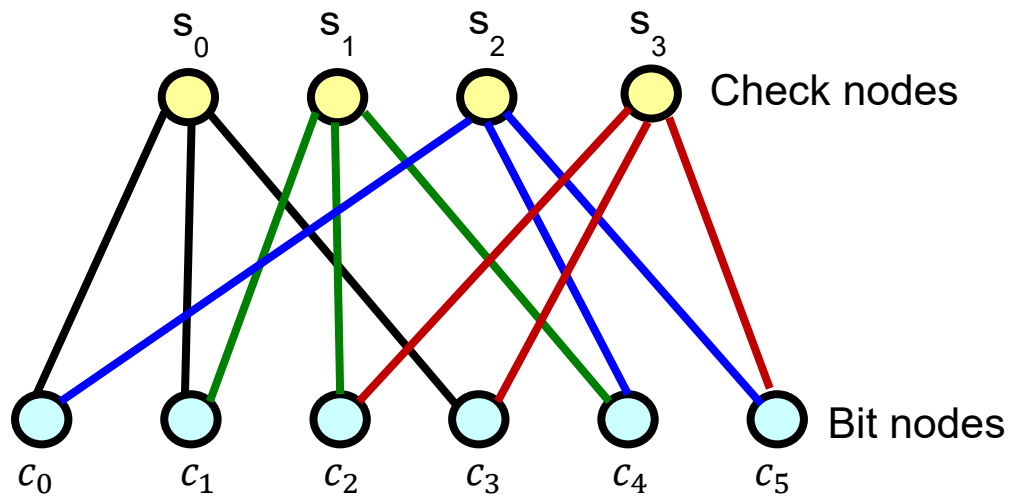


Figure 7



[15 marks]

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